

Testing Manual

A practical, end-to-end guide for validating authentication, shop isolation, inventory, purchasing, sales, payments, returns, reporting, security, and backups.

Application: Multi-Store Retail ERP & POS

Frontend: <http://localhost:3000> **API docs:** <http://localhost:4000/api/docs>

1. Prepare the Test Environment

Start the application with the one-click launcher:

```
start-app.bat
```

Or run it from PowerShell:

```
cd "f:\ERP System"
```

```
.\start-app.bat
```

Default administrator: admin@erp.local / ChangeMeNow!123

- Confirm Docker Desktop is running.
- Confirm PostgreSQL and Redis are available.
- Open the frontend and API documentation URLs above.
- Use a fresh test shop, products, customer, and supplier for repeatable testing.

2. Recommended Test Order

1. Authentication, users, roles, and shops
2. Products and inventory
3. Suppliers and purchases
4. Customers and POS sales
5. Payments
6. Returns and exchanges
7. Transfers between shops
8. Reports
9. Settings, notifications, audit, security, and backup

This order follows the data dependencies of a real retail workflow and makes stock and financial totals easier to reconcile.

3. Authentication and Access

Login and session

1. Open `http://localhost:3000`.
2. Log in with the default administrator account.
3. Refresh the page and confirm the session remains valid.
4. Log out and confirm protected pages are no longer accessible.
5. Try an invalid email and invalid password.

Expected: valid login opens the dashboard; invalid credentials show a clear error; logout clears access.

Users and permissions

- Create a staff user.
- Assign a restricted role.
- Log in as that user.
- Try an allowed operation and a forbidden operation.

Expected: the user can perform only permitted actions and receives a controlled denial elsewhere.

4. Shops and Isolation

1. Create or open two shops, Shop A and Shop B.
2. Assign a user to Shop A only.
3. Create a product or customer in Shop A.
4. Switch to Shop B and search for that record.
5. Try to access Shop A data by changing a URL or request shop identifier.

Expected: shop-scoped records never leak across shops; unauthorized shop access is denied or returns no data.

5. Products and Inventory

Product CRUD

1. Create a product with name, SKU, category, cost, selling price, and quantity.
2. Search by name and SKU.
3. Edit the price and product details.
4. Archive or delete it if the workflow supports that action.

Expected: saved values persist, search is accurate, and duplicate SKUs are rejected.

Inventory movements

- Add stock and verify the quantity.
- Reduce stock with a valid adjustment.
- Attempt a negative quantity or an invalid adjustment.
- Review the inventory ledger and movement history.

Expected: every stock change is recorded, quantities remain accurate, and invalid movements are blocked.

6. Suppliers and Purchases

1. Create a supplier with contact details.
2. Create a purchase using that supplier.
3. Add products, quantities, costs, tax, and discount where supported.
4. Save or receive the purchase.
5. Compare the purchase total with a manual calculation.
6. Verify product stock increased and the ledger contains the movement.

Expected: supplier data is reusable, totals are correct, and receiving stock updates inventory exactly once.

7. Customers, POS, Sales, and Payments

Customer

- Create a customer with name, phone, email, and notes.
- Search using each supported identifier.
- Link the customer to a sale.

POS sale

1. Open the POS screen.
2. Add one product and confirm the line total.
3. Add multiple products and quantities.
4. Apply a discount or tax if available.
5. Select the customer and complete the sale.
6. Review the receipt or sale details.

Expected: subtotal, discount, tax, and grand total are mathematically correct; stock decreases by the sold quantity.

Payment cases

- Cash payment for the exact amount.
- Cash payment with change.
- Card or other configured payment method.
- Partial payment, if supported.
- Invalid or insufficient payment.

Expected: payment status and outstanding balance are correct, and failed payments do not finalize the sale.

8. Returns and Exchanges

Return

1. Select a completed sale.

2. Return part of the order.
3. Return the remaining quantity in a second action if supported.
4. Verify refund amount and returned stock.

Exchange

1. Exchange one sold product for another.
2. Test both an extra amount due and a refund difference.
3. Review the resulting stock movements and payment record.

Expected: returns link to the original sale, quantities cannot exceed the original sale, and stock and money remain reconciled.

9. Transfers Between Shops

1. Start with stock in Shop A.
2. Create a transfer to Shop B.
3. Test draft, sent, received, and cancelled states if available.
4. Receive the transfer in Shop B.
5. Compare both shops' quantities and movement history.

Expected: stock leaves the source only according to the transfer state and arrives at the destination exactly once.

10. Reports

After creating purchases, sales, returns, and transfers, open every report screen and verify:

- sales totals and date filters
- purchase totals
- inventory levels and movement history
- profit or margin calculations
- customer and supplier activity
- shop-specific filtering

Expected: report totals match the transactions created during this test session.

11. Settings, Notifications, Audit, Security, and Backup

Settings

Update company details, currency, tax, locale, and shop settings. Refresh and verify persistence.

Notifications

Trigger a relevant event and verify the notification appears and has the right status.

Audit

Create, edit, and delete a record. Confirm the actor, action, target, and timestamp are recorded.

Security

Test protected routes with no session, expired session, and insufficient permission.

Backup

Start a backup and verify that it completes and creates output in the configured storage location.

12. Negative and Edge-Case Tests

- Submit forms with required fields empty.
- Create a duplicate SKU or duplicate unique identifier.
- Use zero, negative, and excessively large quantities.
- Sell more stock than is available.
- Use invalid decimal amounts and payment totals.
- Refresh during a save operation and retry a submission.
- Use a stale session after logging out in another tab.
- Attempt to view, edit, or delete another shop's data.

Expected: the app shows validation errors, does not silently corrupt data, and returns controlled responses instead of crashes.

13. Final Acceptance Checklist

- ☐ Login, logout, invalid credentials, and session refresh work.
 - ☐ Users, roles, permissions, and shop assignments work.
 - ☐ Cross-shop data access is blocked.
 - ☐ Product creation, editing, searching, and SKU validation work.
 - ☐ Inventory quantities and ledger movements are accurate.
 - ☐ Purchases update stock and totals correctly.
 - ☐ Customers and POS sales work end to end.
 - ☐ Payments, balances, and failed payments behave correctly.
 - ☐ Returns and exchanges reconcile stock and money.
 - ☐ Transfers reconcile source and destination stock.
 - ☐ Reports match manually calculated transactions.
 - ☐ Settings persist and notifications appear.
 - ☐ Audit trail records important actions.
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☐ Backup completes successfully.

14. Suggested Complete Test Session

1. Start the app and log in as administrator.
2. Create two shops and a restricted staff account.
3. Add a supplier and three products.
4. Receive a purchase and verify stock.
5. Create a customer and make a POS sale.
6. Verify payment, receipt, stock reduction, and audit entry.
7. Process a return or exchange.
8. Transfer stock between shops.
9. Review reports and compare totals.
10. Test restricted access and create a backup.